

Math 300 Quiz 5 Spring 2024

1. (8 pts) Use mathematical induction to prove that $8^n - 3^n$ is a multiple of 5 for all integers $n \geq 0$

Over $\rightarrow$

2. (4 pts) Give an example of a function from $\{1, 2, 3, 4\}$ to $\{1, 2, 3\}$ that is neither one-to-one nor onto.

3. (4 pts) Draw a graph or give a rule $[y = f(x)]$ to define a function with domain $\mathbf{R}$ and range $\mathbf{R}$ that is onto but is not one-to-one.

4. (4 pts) State in precise mathematical language what it means for a function f to map set X onto set Y .